SmartLease Edge — Chennai Battle Plan

iQOO Hackathon 2026 · Chennai City Battle · Sep 12-13 2026 · Smart Living track · 30 hours

Companion: `docs/research/IQOO_CHENNAI_2026_RESEARCH_LOG.md` (evidence and provenance for everything asserted here).

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1. Where you actually stand

Three things are true at once, and the plan follows from holding all three.

You have a genuinely strong premise. Four different sensors — camera, microphone, IR emitter, OCR — feeding one signed artifact, entirely offline, with no `INTERNET` permission in the manifest. Most teams at a phone-first hackathon will ship a cloud API with a mobile front end. You are the opposite of that, and it is verifiable rather than claimed.

Your headline model is weak. Mask mAP50 0.267 across four classes; crack recall 0.379. The detector misses roughly six of every ten cracks. This is now measured, not suspected.

Several of your loudest claims are currently false in code. The Hexagon NPU path, GenieX/Llama 3.2 3B, SHA-256, the rupee deduction, dual signatures, ARCore depth. Two of those are being fixed today; the rest cannot be built inside the window.

The winning line is therefore **not** to build more. It is to make a narrower set of claims all of which are true, and to land one live offline run that reaches a rupee figure without stumbling.

2. Decoding the rubric

Criterion	Weight	What actually moves it
End product	30	One complete flow that works when a judge watches it cold
Novelty & impact	20	Functional testing, not visual documentation — "does the AC work", not "does it look fine"
Creative phone use	15	**Device telemetry.** Accrues from using camera / mic / IR, not from arguing
Technical depth	15	Honest numbers with provenance; the ability to answer "how do you know"
Office Kit usage	10	**Device telemetry.** Mirror, clipboard, file transfer, phone-first builds
Demo	10	Reaching the punchline inside the time, on the phone

Two structural facts.

Twenty-five points are telemetry, not opinion. Creative Phone Use plus Office Kit accrue mechanically from how you work during the 30 hours. This is the cheapest quarter of the scoreboard available and needs discipline, not cleverness. Mirror on from hour one. Build on the phone where feasible. Exercise camera, mic and IR in every session rather than only at the end.

Forty points ride on one live run. End Product plus Demo is judged from a flow seen cold. A polished deck in front of a broken demo is the single most reliably punished pattern in every judging source reviewed.

One inference from the research. MLH's published method is stack ranking — judges nominate a top three rather than score each criterion additively (confirmed 2-1). If any of that logic operates here, being memorable to several judges beats being uniformly adequate. One vivid working moment outperforms five half-features. Design the demo around a single thing nobody else will show: a phone that tells you an appliance works, while in airplane mode.

3. Constraints and risks to clear

Pre-existing code — clear this in the first hour. Qualcomm's rules for its own edge-AI events forbid closed-source pre-existing code and require a prior proposal to be "significantly modified" during the event (confirmed 3-0). iQOO/Reskilll's specific wording was not verified. You carry roughly eleven days of pre-event work in a public repo. Ask an organiser directly, and make sure your Round-1 checkpoint (Sat 19:00-22:00) demonstrates substantial work done during the event. This is the only risk on this list that can void everything else.

The IR pattern is your one irreplaceable on-site dependency. The iQOO 15's IR blaster is confirmed present (2-0, official spec page), so the hardware is not in doubt — but the venue AC's actual timing pattern can only be captured in the room. Do it early, not at hour 28. `CommonAcIrProfiles` currently carries carrier frequencies only, and `WalkthroughScreen.kt:191` still prints "placeholder pattern".

The vision model will miss things on camera. With recall 0.379 you cannot point the phone at an arbitrary wall and trust it. Control the scene: rehearse against one specific surface, confirm it fires repeatedly, and bring your own prop if the venue wall does not trigger it. Keep `stain_mould` (AP50 0.092) out of the demo path entirely.

Do not name prior winners. Seven prior-winner claims from the earlier research digest failed verification 0-3. Citing them in front of a sponsor's own engineers is a pure downside bet.

Numbers you may not use. ₹2,400 crore, 1.1 crore agreements, 38% of disputes — no source was ever found. The ">40% of disputes" figure is an industry estimate and must be described as one if challenged. Do not quote 86.7% acoustic accuracy without its interval [0.621, 0.963] over 15 taps.

4. Priority order for the 30 hours

Strictly ordered. Do not start an item before everything above it is done.

1 — Make the money moment real. The demo currently has no ending: there is no rupee logic anywhere in the codebase, yet "₹2,400 deducted from ₹90,000" is the punchline of every slide. Wire `DeductionEngine` into `ReportGenerator`, persist structured `detailJson`, print the balance sheet on the PDF. Nothing else on this list matters as much.

2 — Bank the telemetry 25%. Office Kit mirroring on from the start. Phone-first build sessions. Exercise every sensor in every session.

3 — Capture the venue AC's IR pattern. Early. It gates your most differentiated claim.

4 — Rehearse one flow, timed, at least three times. Airplane mode on, baseline, scan, tap, IR, report. One presenter narrates, one drives. Record a backup video the moment a run succeeds.

5 — Strip every false claim. Delete "<30 ms" and "Hexagon NPU" and "GenieX" from deck, README and spoken script unless wired. The people scoring Technical Depth are the most likely to ask, and a claim that collapses under one question costs more than the claim ever earned.

6 — Only then: NPU or GenieX. Highest effort, highest failure rate, and both are already covered by an honest fallback story. Attempt only with hours to spare.

5. The honesty position — and why it scores

Judges reward candour and punish polished-deck-thin-code; this is the most consistent finding across every judging source reviewed. You have unusually good material for this.

Say plainly: "Our detector gets mask mAP50 0.27 across four classes on mixed public data. That is not good enough to bill someone's deposit on, so the engine refuses to price anything below 60% confidence or anything the colour heuristic flagged — those go to human review. What we assert is the measurement and the signed record, not an infallible detector."

That answer converts your weakest number into evidence of engineering judgement. `DeductionEngine.PRICING_CONFIDENCE_FLOOR` exists in code precisely so this answer is demonstrable rather than rhetorical.

Same move on the fingerprint: it is a SHA-256 integrity digest, **not** a digital signature. There is no keypair, so it proves the record has not changed, not who made it. Saying so before being asked is worth more than the claim you would be defending.

6. Pitch structure (3 minutes, hard stop)

Time	Content
0:00-0:20	One named person, one sentence of pain. "Priya, Anna Nagar, ₹90,000 deposit, twelve WhatsApp photos, no move-in record."
0:20-0:40	The gap: existing tools document how a flat looks. None test whether anything works, and none work offline.
0:40-2:20	**Live demo, airplane mode visible.** Baseline, scan, knuckle tap, IR at the AC, signed report with the balance sheet. Reach the rupee figure by 2:00.
2:20-2:40	Why it can only be this phone: NPU, IR emitter, sensing hub, no network permission in the manifest.
2:40-3:00	Honest status and close. Stop talking and invite questions.

Reach the demo by 40 seconds. Every source reviewed agrees that the demo, not the framing, is what is scored — and a pitch that runs long gets cut before the punchline.

7. Q&A preparation

Question	Answer
What is your accuracy?	Mask mAP50 0.267 across four classes, crack recall 0.379, measured on a 203-image held-out split; `ml/YOLOV8/val_metrics.json`. Which is why the pricing engine refuses low-confidence detections.
Is this a legal document?	It is a tamper-evident record both parties can verify by recomputing the digest. Not a digital signature — no keypair, so it proves integrity, not authorship.
Why not just use ChatGPT and a photo?	It needs network, it uploads a tenant's home to a third party, and it cannot tell you whether the AC works.
Does it run on the NPU?	Today the segmenter runs through PyTorch Lite on CPU. The QNN bundle is exported and in the repo; wiring the Hexagon delegate is the next step, not a claim we are making now.
Where do the repair rates come from?	Team-compiled Chennai contractor estimates, printed as estimates. `RepairTariff.kt` documents the provenance; real deployment replaces them with quotes.
Only 15 tap samples?	Correct, and the interval is wide: 0.867 with 95% CI [0.621, 0.963] against a 0.533 baseline. It beats chance; it is not yet a product claim.

8. Engineering changes made on event day

Change	Status
`deduction/DeductionModels.kt`, `RepairTariff.kt`, `FindingDetail.kt`, `DeductionEngine.kt`	Written
`report/ReportFingerprint.kt` — SHA-256 over canonical record plus file digest	Written
`report/ReportModels.kt` — carries deductions and digest	Edited
Wire into `ReportGenerator` and `WalkthroughScreen`	**Outstanding**
Remove unused `ACCESS_FINE_LOCATION`	**Outstanding**
Delete dead `com/iqoo/multimodal/` package	**Outstanding**
First-ever vision validation metrics	Done, `ml/YOLOV8/val_metrics.json`

Design note: `RepairTariff` uses a per-sq-ft rate with a minimum call-out floor, so a small crack costs ₹1,500 because a contractor charges for the visit — the plan document's figure emerges from real logic instead of being hardcoded.

9. What to hold on to

You will be tempted, somewhere around hour 20, to start wiring the NPU because it sounds impressive. Do not. A working offline flow that ends in a rupee figure, defended with honest numbers, beats a half-wired Hexagon delegate that fails on stage — and the rubric's own weights say so: End Product 30, Technical Depth 15.

The one thing no other team will show is a phone that proves an appliance works while in airplane mode. Build the whole demo around that.

Research Log — iQOO Hackathon 2026, Chennai City Battle

Run date: 2026-09-12 (event day). Method: 5-angle fan-out web search, 26 sources fetched, claims extracted and put to a 3-vote adversarial verification panel (2 of 3 refutes kills a claim).

Run outcome: partial. 79 of 108 agents completed; 29 failed against a session limit, including the final synthesis step. There is therefore no auto-generated cited report — this log is the hand-assembled record of what survived verification, what did not, and what was never answered.

Run ID `wf_e5e44946-90e`. Per-agent transcripts: `.claude/projects/.../subagents/workflows/wf_e5e44946-90e/journal.jsonl`.

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1. Confirmed claims

#	Claim	Vote	Source
C1	The iQOO 15 ships an **infrared blaster** as a listed hardware sensor — the `ConsumerIrManager` 38 kHz appliance demo is hardware-feasible on the target device	2-0	iqoo.com official spec page
C2	Qualcomm's on-site edge-AI rubric is 100 points: **Technical Implementation 40, Use-Case & Innovation 25, Local Processing & Privacy 15, Deployment & Accessibility 10, Presentation & Documentation 10**	3-0	Qualcomm Official Rules PDF (Edge AI Developer Hackathon, New York)
C3	Qualcomm rules **forbid closed-source pre-existing code**; a pre-existing proposal must be "significantly modified to add the AI models"; screening scores whether scope is achievable in the build window	3-0	same PDF
C4	MLH's recommended winner selection is **stack ranking**, not additive scoring: each judge nominates a top three worth 3/2/1 points; winners recur across judges' lists	2-1	MLH judging guide

Why each matters here

- **C1** was the single binary go/no-go in the whole plan. Had the iQOO 15 lacked an IR emitter, an entire demo leg and a rubric differentiator would have died. It is confirmed present.
- **C2** is not this event's rubric, but it is the same sponsor family and shows where silicon vendors put weight: engineering evidence far above pitch polish. iQOO's published weights (End Product 30 / Novelty 20 / Creative Phone Use 15 / Technical Depth 15 / Office Kit 10 / Demo 10) rhyme with it.

- **C3** is the most actionable constraint found. This project carries roughly eleven days of pre-event work in a public repo. Qualcomm's wording would require significant in-event modification. **iQOO/Reskilll's specific rule was not verified** — confirm with an organiser on arrival.
- **C4** changes pitch strategy: if judges nominate a top three rather than score every criterion, being memorable to several judges beats being uniformly competent. One vivid, working moment outperforms five half-features.

2. Refuted claims — do not cite these

All of the following failed verification 0-3 and must not appear in a deck, README, or spoken answer.

Claim	Note
Qualcomm Bengaluru 2025 Grand Prize went to a project because it showed a complete end-to-end on-device pipeline	Refuted 0-3
GameSense (Tech Mahindra) ran Llama 3.2 3B on Hexagon via Qualcomm AI Runtime + GENIE	Refuted 0-3
All five Qualcomm Korea 2025 winners were end-user application products	Refuted 0-3
Korea winners composed off-the-shelf open models rather than training custom ones	Refuted 0-3
Llama-3.2-3B-Instruct was used by two of five Korea winning teams	Refuted 0-3
Winners marketed NPU-local execution as the product value proposition	Refuted 0-3
MLH budgets ~4 minutes per project and enforces a hard 3-minute cap	Refuted 0-3

Important: several of these appear as asserted fact in `~/.claude/skills/hackathon-pitch-deck/references/RESEARCH.md` (section 2). That digest was written 2026-09-06 and its prior-winner claims did not survive re-checking. Whether the sources genuinely fail to support them or the pages were unreachable during verification, the consequence is identical: **do not name prior hackathon winners as precedent.** Staking credibility on an unverifiable claim in front of a sponsor's own engineers is a pure downside bet.

3. Contested (1-2 and 2-1 splits — treat as weak)

- Qualcomm's 40-point Technical Implementation is scored on NPU utilisation, SLM optimisation, latency and energy sub-dimensions (1-2).
- Fully on-device execution is a separately scored 15-point line item (1-2).
- The Windows on Snapdragon hackathon included an explicit "best use of Qualcomm AI Hub models" sponsor-stack criterion (1-2).
- MLH requires a submission video showing a working demo rather than slides (1-2).

Directionally consistent with C2; not solid enough to quote with a number attached.

4. Unanswered — killed by the session limit

The entire device-and-toolchain angle died before voting. None were recovered:

- ARCore device certification and **ARCore Depth API availability on vivo/iQOO** hardware
- Whether Ultralytics ships a first-party QNN exporter, and whether that path yields a `.pte`
- YOLOv8 **Segment** task support on the QNN/ExecuTorch path

- Snapdragon 8 Elite HTP v79 specifics
- Whether Qualcomm publishes a pre-optimised YOLOv8-seg
- iQOO 15 chipset confirmation beyond the spec page

Assessed as non-blocking. Every one sits on the NPU or ARCore branch. The shipping app uses neither: `ArAlignmentTracker` is SensorManager rotation-vector only with no ARCore dependency, and vision inference runs through PyTorch Lite on CPU. Resuming the run to answer them would spend tokens on work that is not happening inside the event window.

5. Also unanswered — no sources found

- **Bengaluru (Aug 29-30) and Pune (Sep 5-6) City Battle winners.** These had already happened and would have been the most directly relevant precedent available. Nothing surfaced.
- Chennai venue specifics, Grand Finale prize split.

6. How to resume, if ever wanted

```
Workflow({
  scriptPath: ".../workflows/scripts/deep-research-wf_e5e44946-90e.js",
  resumeFromRunId: "wf_e5e44946-90e"
})
```

Completed agents replay from cache; only the 29 failures and synthesis re-run. Session limit resets 01:40 IST. **Recommendation: do not.** See section 4.

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7. Local evidence generated during this session

Not from the web — measured here, and more decision-relevant than anything the search returned.

7.1 Vision model validation (new)

`best.pt` validated against the 203-image held-out `unified_defects` validation split. Results committed to `ml/YOLOV8/val_metrics.json`.

Class	Images	Instances	Box mAP50	Mask mAP50	Mask mAP50-95
all	203	785	0.302	0.267	0.136
crack	104	298	0.322	0.276	0.099
peeling	41	120	0.168	0.148	0.085
spalling	85	242	0.585	0.553	0.326
stain_mould	58	125	0.132	0.092	0.034

Crack precision 0.452, **recall 0.379**. Inference 210.7 ms/image on laptop CPU.

Reproduce with an absolute-path data yaml (the committed `data.yaml` uses `path: .`, which ultralytics resolves relative to the wrong directory):

```
python -c "from ultralytics import YOLO; YOLO('best.pt').val(data='<abs>/unified_defects.yaml', imgsz=640, split='val', device='cpu')"
```

Reading of the result. The merged six-dataset corpus spans bridge concrete, pavement and interior walls — domains that do not share visual statistics — and the model is diluted across them. `spalling` is the only strong class and the least relevant to a rental flat. `stain_mould` at AP50 0.092 is effectively non-functional and must stay out of the live demo path. `crack`, the hero class, misses roughly six of every ten instances. The `crack-seg` source alone carried 3,717 images and would likely have produced a far stronger single-class detector; there is no time to retrain inside the event.

This number is why `DeductionEngine` refuses to price any detection below 60% confidence or any heuristic hit. Billing a tenant against a 0.267 mAP50 detector is the attack an ML-literate judge would run, and the architecture should answer it before it is asked.

7.2 Acoustic model (previously recorded)

`ml/models/acoustic_report.json`: LeaveOneGroupOut accuracy 0.867 (LogisticRegression), **95% CI [0.621, 0.963]**, majority baseline 0.533 — over **15 taps from 8 source recordings**. Honest phrasing: beats baseline, sample far too small for a confidence claim. Do not quote 86.7% without the interval.

7.3 Repo state versus the plan document

Audited against `SmartLease_Edge_Complete_App_Structure_and_Team_Matrix.pdf`. Build green (`assembleDebug`, exit 0, 114 MB APK).

Plan claim	Actual	Status
YOLOv8n-Seg INT8 on Hexagon NPU, <30 ms	Ships as `yolov8n_seg.ptl` via **PyTorch Lite on CPU**. `.pte` + `libQnnHtp.so` exist in `ml/YOLOV8/qnn_android_bundle/` but are not in app assets and there is no ExecuTorch dependency	**False**
Llama 3.2 3B via GenieX	`ReportGenerator.synthesizeNarrative()` is a commented placeholder; no GenieX in app code	**False**
SHA-256 fingerprint	Absent; file comment states it is not a cryptographic signature	**Was false — being fixed**
₹2,400 deducted from ₹90,000	No rupee logic anywhere in the codebase	**Was false — being fixed**
Dual on-screen signatures	No signature canvas	**False**
ARCore depth for area math	SensorManager rotation vector only	**False**
Acoustic tap classifier	Real, trained, wired at `WalkthroughScreen.kt:54`	**True**
IR transmit	Real `ConsumerIrManager` wrapper, wired; pattern still placeholder pending on-site capture	**Partly true**
ML Kit OCR, CameraX, Room, offline PDF	Real and wired	**True**
No INTERNET permission	Confirmed absent from manifest	**True**

Also found: `ACCESS_FINE_LOCATION` requested but never used (no reader anywhere in the tree), and a dead second package `app/src/main/java/com/iqoo/multimodal/` carrying its own `MainActivity`.

Changelog

- 2026-09-12: initial log. Partial run (79/108 agents, synthesis failed on session limit). Four confirmed claims; seven prior-winner claims from the 2026-09-06 skill digest refuted 0-3 and retired from use. Added local evidence: first-ever vision validation metrics, acoustic interval, and full claim-versus-code audit.